

Lecture 9 Counting - basic principle, permutations, combinations & partitions

BIO210 Biostatistics

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Aims

- Recap from the math class
- Be familiar with the notations

Principles of counting

- Basic principles of counting
- permutations/ k -permutations
- combinations
- partitions

- Let all outcomes be equally likely
- Then:

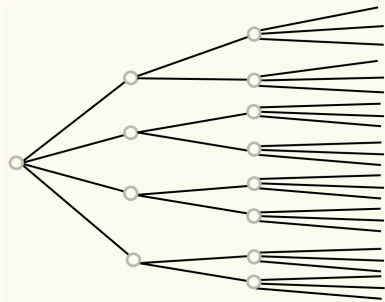
$$\mathbb{P}(A) = \frac{\text{number of outcomes in } A}{\text{total number of outcomes}} = \frac{|A|}{|\Omega|}$$

$|A|, |\Omega|$: cardinality of the set

- All you need to do is: **counting** !

Basic counting principle

- **Basic scenario:** r stages, n_i choices at stage i

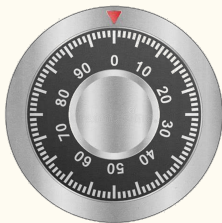


Number of choices is:

$$\prod_{i=1}^r n_i = n_1 \cdot n_2 \cdot n_3 \cdots n_r$$

1. **k -permutations:** Number of ways of ordering k elements chosen from n elements: $n \cdot (n - 1) \cdot (n - 2) \cdots (n - k + 1) = n! / (n - k)!$
2. **Permutations:** Number of ways of ordering n elements is: $n!$
3. Number of subsets of n elements: 2^n

Richard Feynman, The Safecracker



Feynman wanted to brutal force the problem, then

- What is the total number of combinations for a three-number password?
- He found out the dial was not mechanically perfect: ± 2 also works. What is the minimum number of trials to iterate all combinations?
- He noticed people used dates as passwords. Let's assume the year was between 1900-1945. What is the minimum number of trials now?

The Birthday Problem

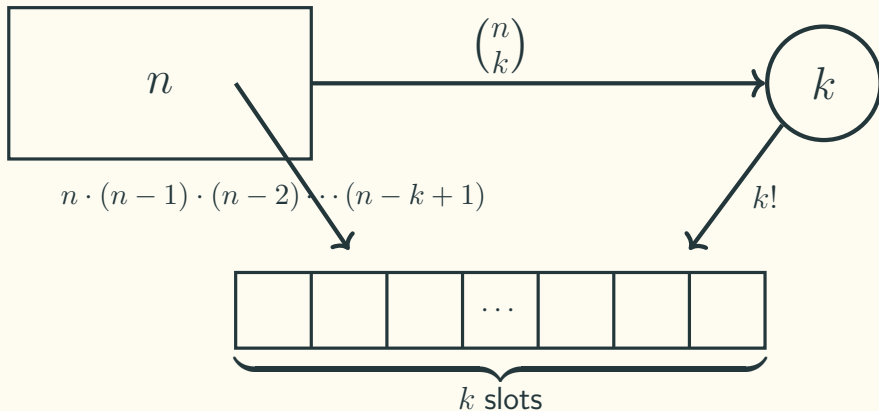
- What is chance that in a group of 25 randomly selected people two or more will be found to share the same birthday?
- $|\Omega| = ?$
- $A = \{ \text{two or more will be found to share the same birthday} \}$. What is $|A|$?



- $A^C = \{ \text{all of them have distinct birthday} \}$. What is $|A^C|$?

Combinations

- Number of k -element subsets of a given n -element set - how many ways are there of choosing k -elements from an n -element set without replacement.

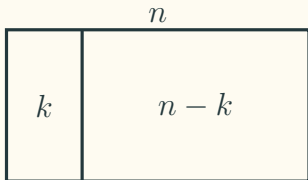


Coin flip example

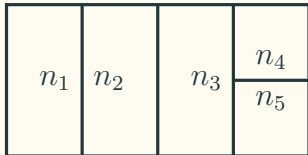
- Event $A = \{ 3 \text{ out of } 10 \text{ independent flips were Hs} \}$
- Given that A has occurred, what is the conditional probability that the first two flips were Hs?



Partitions



$$n = n_1 + n_2 + n_3 + n_4 + n_5$$



$$|\Omega| = \frac{n!}{n_1!n_2!n_3!n_4!n_5!}$$

Example (ABO blood groups): In a room with 100 people, we know that there are 20 people with blood type A, 10 with B, 20 with AB and 50 with O, but we don't have the information of the blood type of each individual person.

1. How many total possible observations are there?
2. If we only know Adam, Bob, Charlie and Dave have four different blood types, how many total possible observations are there for all people in the room?